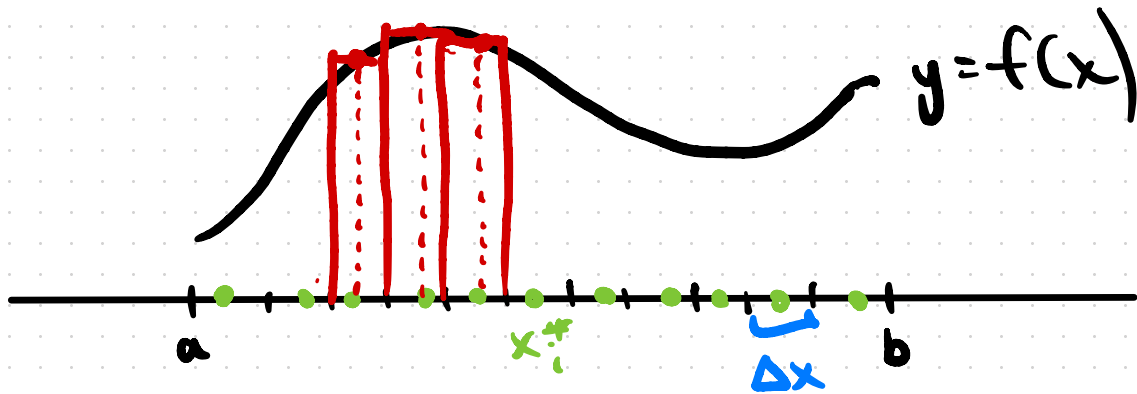


The Definite Integral (5.2)

Suppose we have a function $f(x)$ defined on an interval $[a, b]$, and $f(x)$ is positive on this interval. We can approximate the area under the graph of $f(x)$ by subdividing $[a, b]$ into smaller intervals of length Δx choosing a point x_i^* in each interval, and summing up $f(x_i^*)\Delta x$ over all x_i^* .
 from the i -th interval



The sum of the rectangles is

of intervals $\nearrow$

$$\sum_{i=1}^n f(x_i^*) \Delta x$$

Riemann sum

Consider a small interval $[x_i, x_{i+1}] \leftarrow i^{\text{th}}$ interval

Some possible choices of x_i^* :

- Left endpoint of interval: x_i Area: $f(x_i) \Delta x$
- Right endpoint of interval: x_{i+1} Area: $f(x_{i+1}) \Delta x$
- Midpoint of interval: $\frac{1}{2}(x_i + x_{i+1})$, " $\bar{x}_i$ "

$$\text{Area: } f\left(\frac{x_i + x_{i+1}}{2}\right) \Delta x$$

If we subdivide into smaller and smaller intervals, we get a better approximation. The true area is given by

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

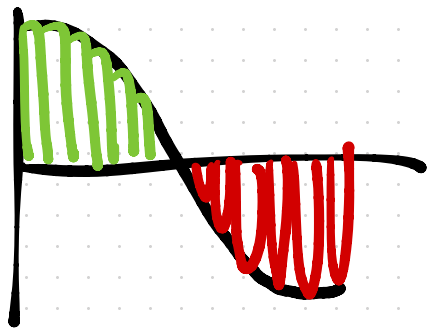
of
intervals $\nearrow$

(Assuming the limit exists.)

If f takes on both positive and negative values, then we can still subdivide $[a, b]$ into intervals of length Δx , choose x_i^* from each interval, and compute

$$\sum_{i=1}^n f(x_i^*) \Delta x$$

Negative values of $f(x_i^*)$ will contribute negatively to the sum.



$$\sum_{i=1}^n f(x_i^*) \Delta x = \left(\begin{array}{l} \text{Area of the rectangles} \\ \text{above } x\text{-axis} \end{array} \right) - \left(\begin{array}{l} \text{Area of the rectangles} \\ \text{below } x\text{-axis} \end{array} \right)$$

Definition: Let f be a function defined on an interval $[a, b]$. The **definite integral** of f from a to b is

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x \quad \left(\begin{array}{l} \text{assuming this} \\ \text{limit exists} \end{array} \right)$$

where $\Delta x = \frac{b-a}{n}$ (length of a small interval), and x_i^* is chosen from the i^{th} interval. We denote the definite integral by

$$\int_a^b f(x) dx$$

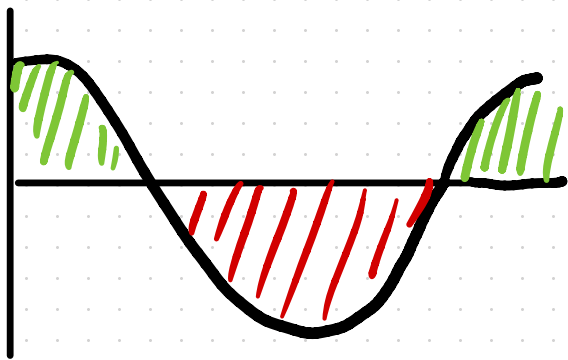
"the integral of f
from a to b "

Graphically,

$$\int_a^b f(x) dx$$

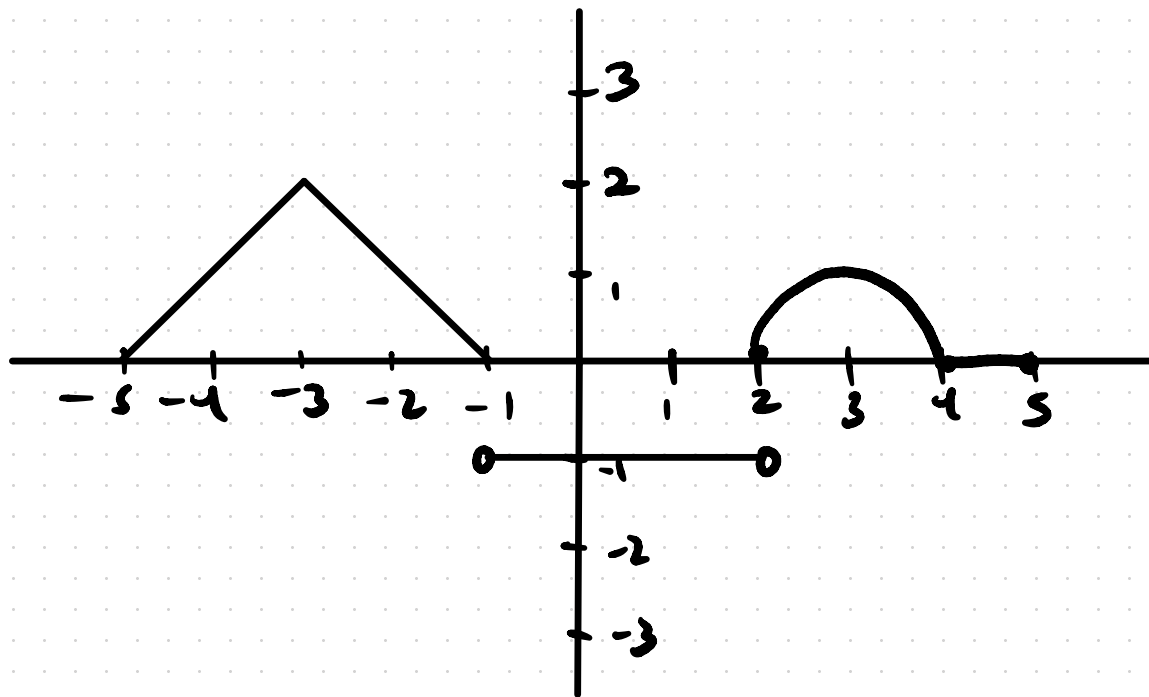
is the

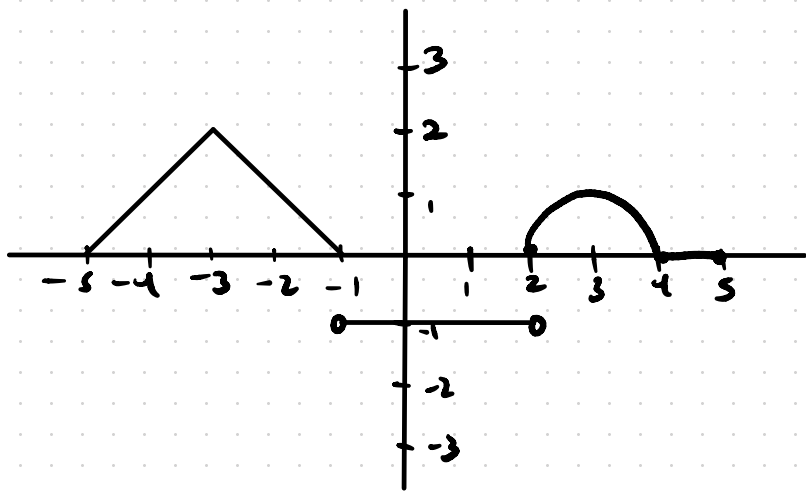
$$\left(\begin{array}{c} \text{area under } f \\ \text{above } x\text{-axis} \end{array} \right) - \left(\begin{array}{c} \text{area above } f \\ \text{under } x\text{-axis} \end{array} \right)$$



$$\int_a^b f(x) dx = \text{green area} - \text{red area}$$

Let f be the following function.





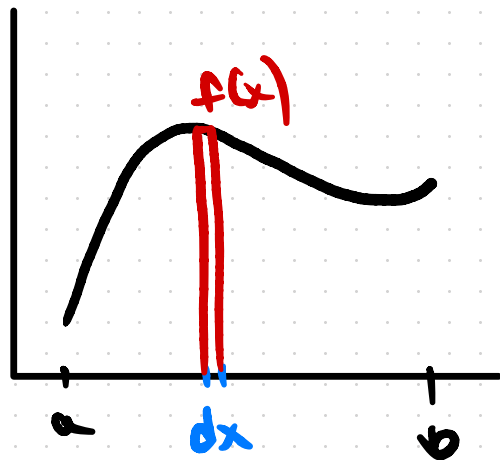
$$\int_{-5}^{-1} f(x) dx = \frac{1}{2} \cdot 4 \cdot 2 = 4$$

$$\int_{-5}^1 f(x) dx = 4 - 2 = 2$$

$$\int_{-1}^4 f(x) dx = -3 + \frac{\pi}{2}$$

$$\int_4^5 f(x) dx = 0$$

Why the funny notation?



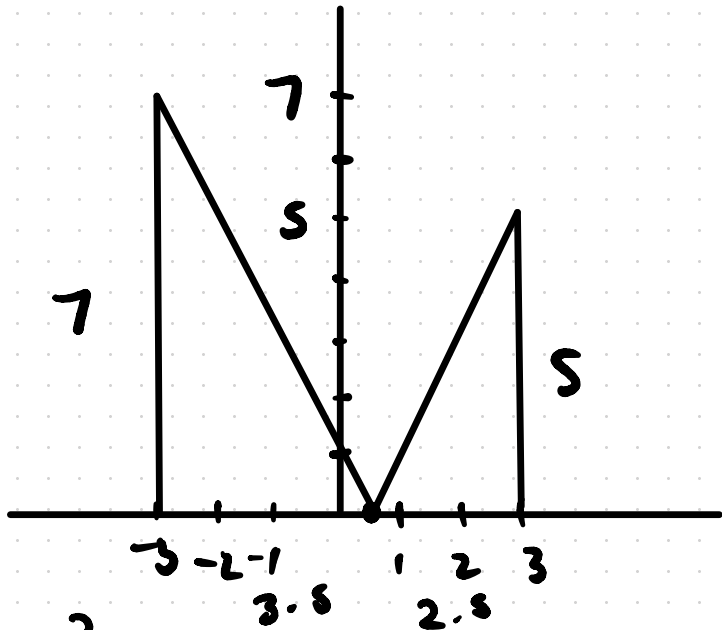
$$\int_a^b f(x) dx$$
$$\sum_{i=1}^n f(x_i^*) \Delta x$$

Arrows point from the summation notation to the integral notation, indicating the limit process.

infinitely small width, but not 0

Evaluate

$$\int_{-3}^3 |2x-1| dx$$



$$2x-1 \geq 0 \text{ when}$$

$$x \geq \frac{1}{2}$$

$$2x-1 \leq 0 \text{ when}$$

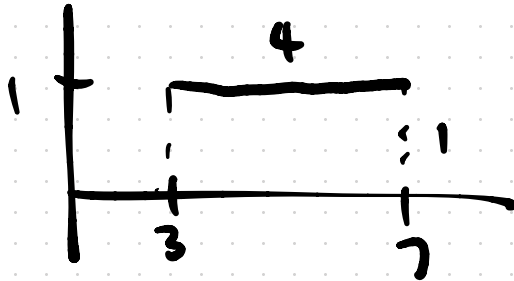
$$x \leq \frac{1}{2}$$

$$\int_{-3}^3 |2x-1| dx = \frac{1}{2} \cdot 7 \cdot \frac{7}{2} + \frac{1}{2} \cdot 5 \cdot \frac{5}{2} = \frac{74}{4}$$

Evaluate

$$\int_3^7 dx$$

$$\int_3^7 1 \cdot dx$$

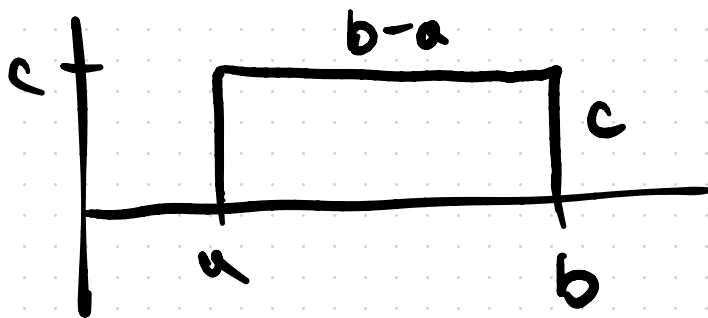


$$\int_3^7 dx = 4$$

In general,

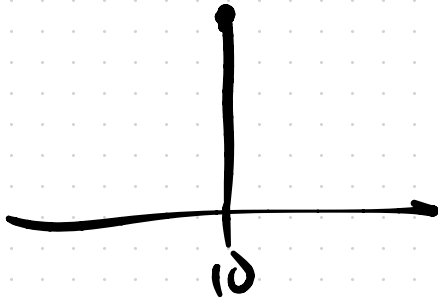
$$\int_a^b c dx = c(b-a)$$

$\swarrow$ c a constant



Evaluate

$$\int_{10}^{10} \frac{x^3 \sin^4 x}{\sqrt{\ln x + \tan x}} dx$$



"
0

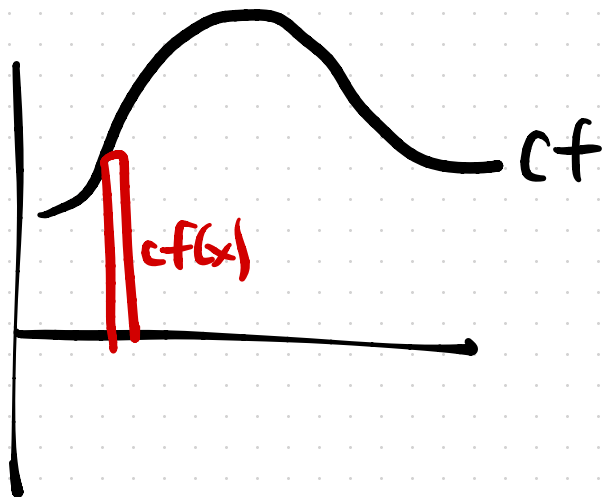
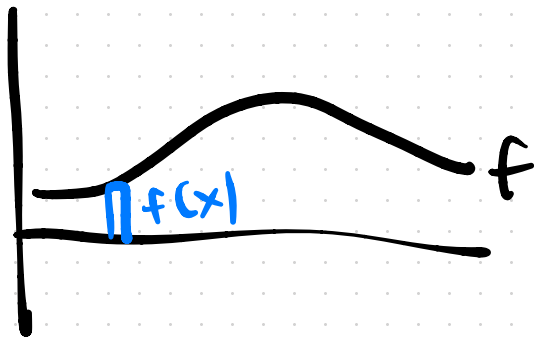
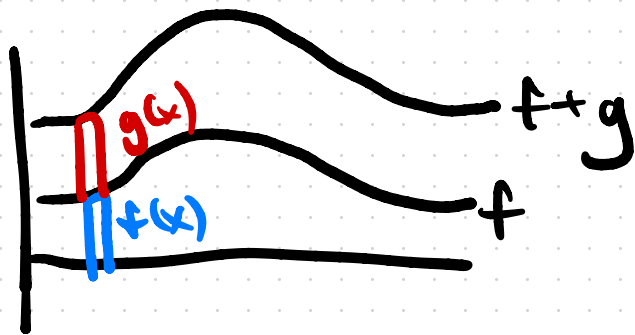
Basic properties of definite integrals

$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

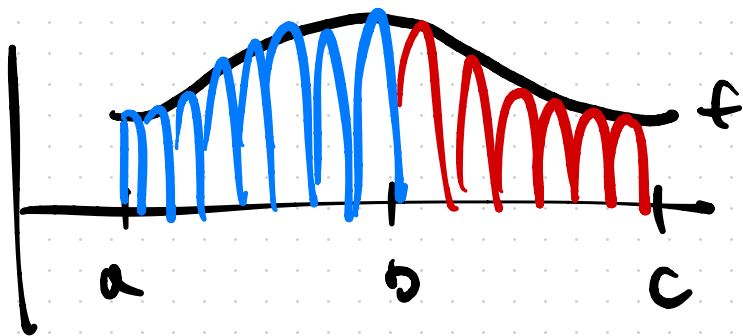
$$\int_a^b (f(x) - g(x)) dx = \int_a^b f(x) dx - \int_a^b g(x) dx$$

$$\int_a^b c f(x) dx = c \int_a^b f(x) dx$$

c a constant

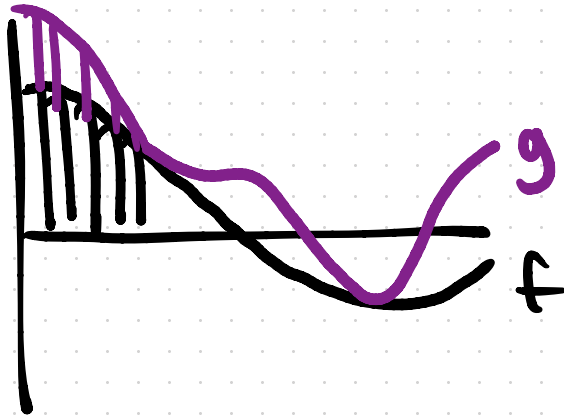


$$\int_a^b f(x)dx + \int_b^c f(x)dx = \int_a^c f(x)dx$$



If $f(x) \leq g(x)$ for all $a \leq x \leq b$, then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx$$



Prove that

$$0 \leq \int_{-100}^{100} (\sin x + 1) dx \leq 400$$

Know that $-1 \leq \sin x \leq 1$

Therefore $0 \leq \sin x + 1 \leq 2$

$$\begin{aligned} \int_{-100}^{100} 0 dx &\leq \int_{-100}^{100} (\sin x + 1) dx \leq \int_{-100}^{100} 2 dx \\ 0 &\leq \qquad \qquad \qquad \leq 400 \end{aligned}$$